HiSem: Hierarchical Semantic Disentangling for Remote Sensing Image Change Captioning

Man Wang, Chenyang Liu, Wenjun Li, Feng Ni, Bing Jia, *Member, IEEE*, Baoqi Huang, Riting Xia, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*—Remote sensing image change captioning (RSICC) aims to achieve high-level semantic understanding of genuine changes occurring between bitemporal images. Despite notable progress, existing methods are fundamentally limited by a shared modeling assumption: changed and unchanged image pairs, which have intrinsically different semantic granularities, are processed under a unified modeling strategy. This modeling inconsistency leads to semantic entanglement between coarse-grained change existence judgment and fine-grained semantic understanding. To address the above limitation, we propose a novel hierarchical semantic disentanglement network (HiSem) that explicitly disentangles semantic representations of different granularities. Specifically, we first introduce the bidirectional differential attention modulation (BDAM) module that leverages discrepancy-aware attention to enhance cross-temporal interactions, thereby amplifying true change signals while suppressing irrelevant variations. Building upon this, we design a hierarchical adaptive semantic disentanglement (HASD) module that performs adaptive routing at two hierarchical levels: a coarse-grained image-level routing mechanism distinguishes changed and unchanged image pairs, while a fine-grained token-level mixture-of-experts (MoE) block models diverse and heterogeneous change semantics for changed samples. Extensive experiments on two benchmark datasets demonstrate that HiSem outperforms previous methods, achieving a significant improvement of +7.52% BLEU-4 on the WHU-CDC dataset. More importantly, our approach provides a structured perspective for RSICC by explicitly aligning model design with the intrinsic semantic heterogeneity of bitemporal scenes. The code will be available at: https://github.com/Man-Wang-star/HiSem**

***Index Terms*—Change captioning, hierarchical semantic disentanglement, remote sensing, Transformer.**

I. Introduction

REMOTE sensing image change captioning (RSICC) is an emerging vision-language task in the remote sensing community. Given a pair of remote sensing images (RSIs) capturing the same geographical area at different times, the RSICC task aims to automatically describe the differences between bitemporal images in natural language [1], [2]. RSICC technology plays an important role in various fields, such as land planning [3], [4], [5], urbanization analysis [6], [7], and environmental monitoring [8], [9].

RSICC can be viewed as a natural evolution of remote sensing change analysis toward richer semantic understanding. Early binary change detection (BCD) methods mainly focus on pixel-level localization, answering where changes occur while largely ignoring their semantic meaning [10], [11], [12]. Semantic change detection (SCD) further introduces category-level supervision to identify what type of objects have changed, yet it still remains primarily a visual classification task limited to predefined semantic labels [13], [14], [15]. RSICC pushes this paradigm one step further by introducing natural language as the output modality. Instead of simply identifying change regions or categories, RSICC aims to generate human-readable descriptions that explicitly explain what has changed and how these changes manifest over time, including object attributes, spatial relationships, and temporal evolution [16], [17], [18].

Most existing RSICC methods are built upon an encoder–decoder framework and generally follow a three-stage paradigm: bitemporal visual representation, change perception, and language generation [16], [17], [18], [19], [20], [21], as illustrated in Fig. 1(a). In the visual representation stage, Siamese pretrained backbone networks, such as convolutional neural networks (CNNs) [22] or Vision Transformers (ViTs) [23], are commonly employed to extract spatially structured and semantically rich features from bitemporal images. In the change perception stage, dedicated interaction mechanisms are introduced to explicitly model cross-temporal relationships between paired features, enabling the model to highlight semantically meaningful differences while suppressing irrelevant variations. Finally, language models (e.g., Transformer

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Man Wang, Bing Jia, Baoqi Huang, and Riting Xia are with the College of Computer Science and the Research Center for Spatiotemporal Intelligence, Inner Mongolia University, Hohhot 010021, China (e-mail: xiart@imu.edu.cn).

Chenyang Liu is with the Department of Aerospace Intelligent Science and Technology, School of Astronautics, the State Key Laboratory of Virtual Reality Technology and Systems, and the Shen Yuan Honors College, Beihang University, Beijing 100191, China.

Wenjun Li and Feng Ni are with China Mobile Communications Group Inner Mongolia Company Ltd., Hohhot 010010, China.

Zhenwei Shi is with the College of Computer Science and the Research Center for Spatiotemporal Intelligence, Inner Mongolia University, Hohhot 010021, China, and also with the Department of Aerospace Intelligent Science and Technology, School of Astronautics, and the State Key Laboratory of Virtual Reality Technology and Systems, Beihang University, Beijing 100191, China (e-mail: shizhenwei@imu.edu.cn).

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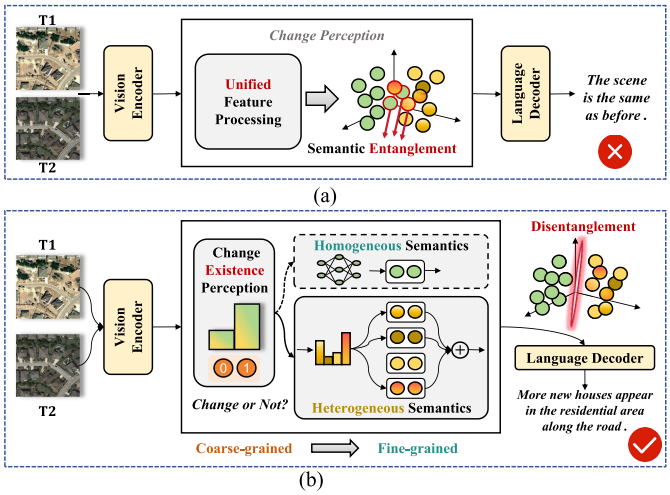


Fig. 1. (a) Previous methods uniformly process changed and unchanged image pairs, resulting in semantic entanglement. (b) Our method explicitly distinguishes between coarse-grained existence perception and fine-grained semantic understanding.

[24], LSTM [25], and Mamba [26]) map the resulting change-aware visual representations into coherent natural language.

Within this framework, the quality of change-aware representations learned during the change perception stage largely determines the accuracy of the generated captions. However, remote sensing temporal scenes are inherently complex and often contain numerous irrelevant variations caused by illumination differences, seasonal effects, or sensor noise. These factors introduce two intertwined challenges for RSICC: 1) suppressing irrelevant variations to accurately determine whether changes exist and where they occur and 2) extracting fine-grained semantic information from true change regions to support coherent and informative language generation. To address these challenges, most existing studies focus on improving the change perception stage. For example, Li et al. [27] proposed a cross-temporal attention (CTA) mechanism to model direct interactions between concatenated bitemporal features, strengthening salient change cues while suppressing background interference. To further enrich semantic modeling, Sun et al. [28] constructed a scene graph based on difference features and employed a graph convolutional network (GCN) to capture object attributes and interobject relationships within change regions.

Despite these advances, existing methods still suffer from a fundamental limitation. Most approaches adopt a unified modeling strategy for both changed and unchanged image pairs, implicitly assuming that they can be processed at the same semantic granularity. In practice, however, unchanged image pairs usually correspond to simple descriptions (e.g., "no change has occurred"), whereas changed image pairs require modeling diverse and fine-grained semantics (e.g., "a building is constructed adjacent to the road"). Treating these two fundamentally different cases in a uniform modeling manner leads to semantic entanglement between coarse-grained change existence judgment and fine-grained change semantic understanding. This observation motivates the need for a more structured change-aware modeling approach that explicitly disentangles semantic representations across different granularities.

To address the above limitation, we propose a hierarchical semantic disentanglement network (HiSem) that explicitly disentangles semantic representations of different granularities, as illustrated in Fig. 1(b). Specifically, we first employ a ViT of the pretrained CLIP model [29] to extract semantic representations for bitemporal images. To obtain more discriminative bitemporal features, we introduce a bidirectional differential attention modulation (BDAM) module, which leverages bitemporal difference features as attention bias to explicitly modulate cross-temporal feature interactions in both "before-to-after" and "after-to-before" directions. This design enhances bitemporal features of genuinely changed regions while suppressing irrelevant temporal variations.

Furthermore, to achieve semantic disentanglement between changed and unchanged image pairs, we propose a hierarchical adaptive semantic disentanglement (HASD) module that separately models their semantics. At the coarse-grained level of change existence perception, an image-level routing mechanism is employed to determine whether an image pair is changed or unchanged based on enhanced difference features, and to adaptively select corresponding processing paths. At the fine-grained semantic modeling level, changed image pairs, which exhibit complex and diverse semantic changes, are processed by a mixture-of-experts (MoE)-based block, where each expert specializes in capturing distinct implicit change semantics. In contrast, unchanged image pairs, whose semantics are relatively homogeneous and simple, are routed through a lightweight block to extract coarse-grained invariant representations, thereby preserving computational efficiency. Finally, the disentangled visual representations are fed into a Transformer-based decoder to generate natural language descriptions that faithfully reflect the temporal semantic states between bitemporal images.

By aligning model design with the intrinsic semantic heterogeneity of bitemporal scenes, HiSem provides a structured solution that bridges coarse-grained change existence perception and fine-grained semantic understanding, enabling more accurate and interpretable change captioning.

Our contributions are summarized as follows.

1) Unlike previous methods that uniformly process changed and unchanged image pairs, resulting in semantic entanglement, we propose a hierarchical semantic disentangling model that distinguishes between coarse-grained change existence perception and fine-grained semantic understanding to alleviate the semantic entanglement.
2) To achieve semantic disentanglement, the BDAM module leverages bitemporal difference features as attention bias to guide cross-temporal feature interactions and enhance discriminative change cues, while the HASD module performs coarse-to-fine semantic routing to adaptively model changed and unchanged samples with different semantic granularities.
3) Extensive experiments on two large benchmark datasets demonstrate that HiSem consistently outperforms previous methods with a significant margin (+7.52% BLEU-4

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on WHU-CDC), validating the effectiveness of the proposed hierarchical semantic disentanglement approach.

II. Related Works

RSICC has emerged as an important task in the intelligent interpretation of remote sensing imagery [30], [31], [32], [33] and has attracted increasing research attention in recent years. Most existing methods follow a three-stage paradigm: 1) bitemporal visual representation; 2) change perception; and 3) language generation.

A. Bitemporal Visual Representation

In the first stage, Siamese or dual-stream architectures with CNNs or ViTs pretrained on large-scale natural image datasets are commonly used to extract generic spatial and semantic features from bitemporal images [1], [17], [34]. Considering the inherent scale variability in remote sensing imagery, multiscale feature extraction is further introduced to capture changes at different spatial resolutions, where methods based on ResNet101 [18], [22], [35] exploit hierarchical CNN features, while Transformer-based encoders such as SegFormer [36], [37], [38], [39] provide multiscale representations with enhanced global context modeling.

To mitigate the domain gap between natural image pretraining and downstream bitemporal remote sensing imagery, Zhou et al. [16] proposed a single-stream extractor network (SEN) that applies contrastive learning to concatenated bitemporal RSIs, aiming to improve feature transferability while maintaining computational efficiency. Furthermore, Yang et al. [40] introduced a hierarchical multiattentive network (MADiffCC) integrated with a diffusion model, which is pretrained on an RSI dataset to effectively model data distributions and temporal–spatial contextual information.

B. Change Perception

The change perception stage aims to learn discriminative representations for actual changes while suppressing irrelevant variations such as illumination differences, seasonal effects, or sensor noise. From a methodological perspective, prevailing research focuses on optimizing bitemporal feature fusion, which can be systematically categorized into spatial-level, temporal-level, and joint spatio-temporal fusion.

Early research primarily emphasizes spatial-level fusion, modeling discrepancies between bitemporal feature maps to achieve precise change localization. RSICCformer [17] bolsters change-aware representations by utilizing cross-temporal feature differences as keys to modulate original features. Diverging from this explicit difference-guided approach, Chg2Cap [1] adopts an alternative strategy that first refines individual temporal embeddings to capture intrainformation, subsequently integrating them via concatenation and stacked attention layers to model intertemporal correlations. SCNet [41] further highlights the importance of channelwise dependencies, introducing a spatial–channel attention encoder to sequentially capture interregion relationships and semantic attributes. MAF-Net [42] aggregates paired hierarchical bitemporal features through a multiscale change aware encoder, utilizing Transformer-based difference and content cross attention to capture discriminative change information across diverse scales. While effective at capturing spatial cues, these approaches remain limited in modeling temporal evolution.

To address temporal dependencies, MFRNet [43] adopts a joint attention mechanism that utilizes concatenated bitemporal features as queries to mutually guide attention distributions across temporal branches. Li et al. [27] enhanced temporal modeling by explicitly constraining the consistency between "before-to-after" and "after-to-before" transitions. This alignment of symmetrical change features is achieved via a bidirectional triplet ranking loss, which is incorporated to enforce symmetry in the latent space. In contrast to direct feature differencing, Ozelbas et al. [44] modeled temporal variation by decoupling change direction and magnitude and used temporal gated fusion to regulate the contributions of preevent and postevent features within a change-aware memory for decoding.

Despite these advances, many methods still neglect the intrinsic correlations between spatial and temporal cues, yielding incoherent spatio-temporal representations. RSCaMa [45] addresses this by leveraging the Mamba architecture and state-space models (SSMs) to jointly model spatial textures and temporal dynamics within a unified representation. To mitigate the sensitivity of spatio-temporal features to environmental variations, FST-Net [38] further incorporates frequency-domain decomposition into the joint modeling process. More recently, BTCChat [46] extends explicit spatio-temporal fusion to multimodal change understanding, introducing a change extraction module that reinforces fine-grained local spatio-temporal correlations by adding cosine-similarity embeddings between bitemporal features to their concatenated visual representation. SACNet [47] further advances change perception by refining CLIP-based semantic representations with CNN-derived spatial discrepancies and deriving change response masks from intertemporal feature dissimilarity, which jointly promotes spatial-semantic alignment and suppresses redundant background responses.

From the perspective of auxiliary information integration, several studies leverage change detection to enhance change perception in RSICC. For instance, Pix4Cap [48] incorporates an auxiliary change detection branch supervised by pseudo-labels from the pretrained change detection model to provide spatial constraints for change localization. Furthermore, Semantic-CC [49] incorporates foundational semantic priors from the segment anything model (SAM) [50] via a multitask semantic aggregation neck to mediate the interaction between change detection and RSICC features, facilitating pixel-level semantic guidance for sentence generation. Whereas Li et al. [51] directly leveraged the P2V-CD [52] model to generate explicit change masks, which are elementwise multiplied with the original images to suppress task-irrelevant background noise while preserving critical spatial details.

Despite these advances, most methods still process changed and unchanged image pairs at a uniform semantic granularity, leading to potential semantic entanglement between coarse-grained change existence and fine-grained change

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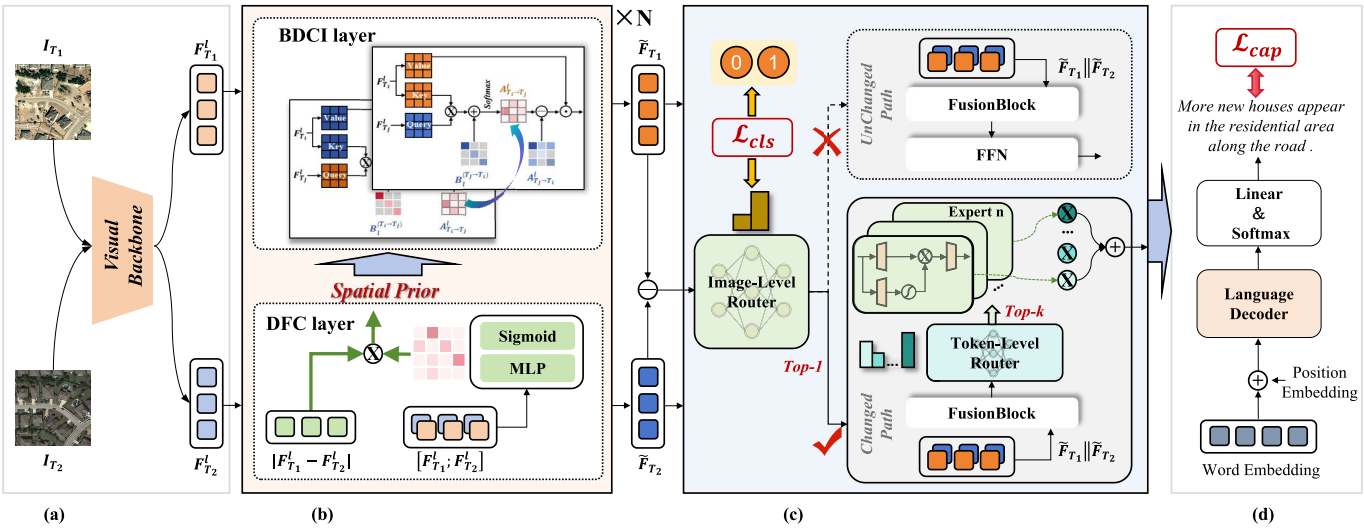


Fig. 2. Overview of our proposed HiSem, which consists of four components: (a) visual encoder, (b) BDAM, (c) HASD, and (d) language decoder.

semantics. To address this limitation, our HiSem network explicitly disentangles semantic representations across two levels: coarse-grained change identification via the BDAM module, which guides cross-temporal interactions using spatial priors, and fine-grained modeling of heterogeneous change patterns through the HASD module, which leverages MoE-based experts to adaptively capture diverse change semantics.

C. Language Generation

In the language generation stage, language models (e.g., LSTM [25], Transformer [24], and Mamba [26]) map the change-aware visual representations into coherent natural language descriptions. Early methods primarily relied on recurrent neural networks (RNNs) or support vector machines (SVMs) for sequential description generation [53], [54]. With the rapid development of deep learning, Transformer-based language decoders have become the predominant choice [1], [17], [34], [55], due to their strong capability in modeling cross-modal dependencies via cross-attention mechanisms. To further improve the quality of generated descriptions, several studies focus on improving vision-language attention mechanisms to enable more effective cross-modal interaction [34], [40], [56]. For example, MADiffCC [40] introduces a gated multihead cross-attention (GMCA) mechanism to adaptively select and aggregate essential multiscale features for enhanced change description generation. However, the quadratic complexity of Transformers limits their scalability. Accordingly, some studies explore more efficient SSMs-based decoders [26], [57], [58] with linear complexity. A representative example is Liu et al. [45], which explores the Mamba-based decoder for efficient caption generation.

More recently, large language models (LLMs) have been introduced to RSICC to improve semantic understanding and caption generation. Existing studies can be broadly categorized into two lines according to their adaptation strategies: prompt or instruction-based adaptation and parameter-efficient fine-tuning (PEFT). Prompt-based methods aim to adapt LLMs without updating their parameters. Prompt-CC [2] introduces a multiprompt learning strategy to condition pretrained LLMs on image-level and class-specific cues, enabling effective change-aware caption generation via conditional prompting. KCFI [21] utilizes key change features as visual instructions into align visual and linguistic modalities, enabling the frozen LLMs to generate more accurate change-aware descriptions through task-specific fine-tuning. Furthermore, PEFT approaches have been explored to balance performance and efficiency, including low-rank adaptation (LoRA)-based methods [59], [60], [61], [62], [63] and adapter-based methods [64], [65]. These approaches introduce a small number of trainable parameters while keeping the original parameters of LLMs frozen, enabling efficient adaptation to RSICC tasks. However, such over-parameterized LLMs often impose substantial computational overhead.

III. Methodology

A. Overview of HiSem

As illustrated in Fig. 2, the proposed HiSem network consists of four core components: a visual encoder, a BDAM module, a HASD module, and a language decoder. Given a bitemporal image pair (I_{T1}, I_{T2}), we first employ a CLIP-based vision encoder pretrained on large-scale image–text data to extract semantically rich representations [29]. The resulting visual embeddings are denoted as $\mathbf{F}_{T_1}, \mathbf{F}_{T_2} \in \mathbb{R}^{B\times L\times D}$, where B is the batch size, L is the number of visual tokens, and D is the feature dimension. Positional embeddings are added to preserve spatial structure prior to further processing.

The BDAM module then refines these visual embeddings to enhance change-sensitive regions. It comprises two sequential sublayers per layer: 1) a discrepancy-aware feature conditioning (DFC) sublayer that highlights semantically informative difference channels and 2) a bidirectional discrepancy-guided cross-temporal interaction (BDCI) sublayer that explicitly modulates cross-temporal feature interactions in both "before-to-after" and "after-to-before" directions using learnable difference priors. This design strengthens genuine change features while suppressing irrelevant temporal variations.

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Algorithm 1 Overall Procedure of the Proposed HiSem

Require: Bi-temporal remote sensing image pair $I = (I_{T_1}, I_{T_2})$.
Ensure: Generated change caption $\mathcal{Y}$.

Definitions:
R_I: coarse-grained image-level routing.
Step 1: Feature extraction
$\mathbf{F}_{T_1}^0 \leftarrow Backbone(I_{T_1})$
$\mathbf{F}_{T_2}^0 \leftarrow Backbone(I_{T_2})$
Step 2: Bidirectional differential attention modulation
for $l = 1$ to N **do**
  $(\hat{\mathbf{F}}_{T_1}^l, \hat{\mathbf{F}}_{T_2}^l) \leftarrow DFC^l(\mathbf{F}_{T_1}^{l-1}, \mathbf{F}_{T_2}^{l-1})$
  $(\tilde{\mathbf{F}}_{T_1}^l, \tilde{\mathbf{F}}_{T_2}^l) \leftarrow BDCI^l(\hat{\mathbf{F}}_{T_1}^l, \hat{\mathbf{F}}_{T_2}^l)$
end for
Step 3: Hierarchical adaptive semantic disentanglement
$(\mathcal{E}_k, \alpha_k) \leftarrow R_I(\tilde{\mathbf{F}}_{T_2}^N - \tilde{\mathbf{F}}_{T_1}^N)$
if $\mathcal{E}_k = 0$ **then**
  Unchanged semantic path
  $\mathbf{F}_H^0 \leftarrow FusionBlock_{\mathcal{E}_0}(\tilde{\mathbf{F}}_{T_1}^N, \tilde{\mathbf{F}}_{T_2}^N)$
  $\mathbf{Y} \leftarrow FFN_{\mathcal{E}_0}(\mathbf{F}_H^0)$
else
  Changed semantic path
  $\mathbf{F}_H^1 \leftarrow FusionBlock_{\mathcal{E}_1}(\tilde{\mathbf{F}}_{T_1}^N, \tilde{\mathbf{F}}_{T_2}^N)$
  $\mathbf{Y} \leftarrow MoEBlock_{\mathcal{E}_1}(\mathbf{F}_H^1)$
end if
Step 4: Caption decoding
$\mathcal{Y} \leftarrow TransformerDecoder(\mathbf{Y})$
return $\mathcal{Y}$

Following BDAM, the HASD module performs hierarchical semantic disentanglement. At the coarse level, an image-level routing mechanism classifies pairs as changed or unchanged, directing them into corresponding expert paths. Unchanged pairs are processed by a lightweight block to extract invariant semantics efficiently, whereas changed pairs are routed to an MoE-based block to capture diverse and fine-grained change semantics. Finally, the disentangled visual representations are fed into a Transformer-based language decoder, generating natural language descriptions that faithfully reflect temporal semantic information between bitemporal images.

The overall procedure of the proposed HiSem model is shown in Algorithm 1.

B. Bidirectional Differential Attention Modulation

As illustrated in Fig. 3, the proposed BDAM module is designed to progressively refine bitemporal representations by explicitly incorporating difference-driven cues. Each BDAM layer follows a spatial-to-temporal progressive modulation strategy, consisting of two sequential sublayers: 1) DFC for spatially localized change enhancement and 2) BDCI for modeling directional change dependencies.

1) DFC Sublayer: The first sublayer aims to enhance change-sensitive responses while suppressing semantically irrelevant variations caused by illumination differences, seasonal effects, or sensor noise. Given bitemporal embeddings

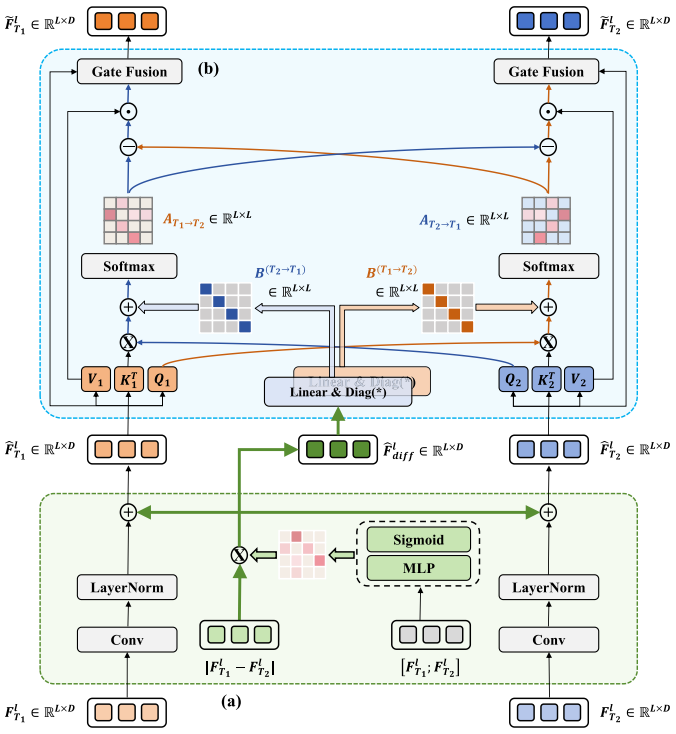


Fig. 3. Structure of the BDAM. (a) DFC sublayer reweights the aggregated bitemporal features to spatially enhance change-sensitive responses. (b) BDCI sublayer exploits the difference features as learnable attention cues to explicitly modulate cross-temporal interactions, capturing temporal dependencies between bitemporal images.

$\mathbf{F}_{T_1}^l, \mathbf{F}_{T_2}^l \in \mathbb{R}^{L\times D}$ at the lth BDAM layer, we first compute absolute difference features

$$\mathbf{F}_{\text{diff}}^l = \left|\mathbf{F}_{T_1}^l - \mathbf{F}_{T_2}^l\right|. \tag{1}$$

Rather than directly using these raw differences, we introduce a lightweight conditioning mechanism to adaptively reweight feature channels according to their relevance to change perception. Specifically, a two-layer MLP generates channelwise modulation weights from concatenated bitemporal features

$$\mathbf{G}_{\text{cond}}^l = \sigma\left(\mathbf{W}_2\phi\left(\mathbf{W}_1\left[\mathbf{F}_{T_1}^l; \mathbf{F}_{T_2}^l\right]\right)\right) \tag{2}$$

$$\hat{\mathbf{F}}_{\text{diff}}^l = \mathbf{F}_{\text{diff}}^l \odot \mathbf{G}_{\text{cond}}^l \tag{3}$$

where $[\cdot;\cdot]$ denotes the channelwise concatenation, $\phi(\cdot)$ and $\sigma(\cdot)$ represent the ReLU and sigmoid activations, respectively, and $\odot$ denotes the elementwise multiplication.

To preserve the local spatial structure, each temporal feature map is reshaped into its 2-D form and processed using convolution. The resulting features are then conditioned by the reweighted difference representations $\hat{\mathbf{F}}_{\text{diff}}^l$, yielding spatially enhanced bitemporal features $\hat{\mathbf{F}}_{T_1}^l$ and $\hat{\mathbf{F}}_{T_2}^l$. This sublayer provides stable and localized discrepancy cues, which serve as a foundation for subsequent cross-temporal interaction.

2) BDCI Sublayer: While the previous sublayer enhances local discrepancy cues, effective change modeling further requires capturing temporal dependencies between bitemporal features. To this end, we introduce a BDCI mechanism.

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Specifically, we first leverage the discrepancy-enhanced features $\hat{\mathbf{F}}^{l}_{\text{diff}}$ to construct a structured, positionwise attention bias

$$\mathbf{B}^{(d)} = \alpha^{(d)} \cdot \text{Diag}\left(\mathbf{W}^{(d)}\hat{\mathbf{F}}^{l}_{\text{diff}}\right), \quad d \in \{T_1 \to T_2,\ T_2 \to T_1\} \quad (4)$$

where $\mathbf{W}^{(d)}$ is the learnable projection that aggregates channel-wise discrepancy responses into scalar positional scores, $\alpha^{(d)}$ is the learnable scaling factor, and Diag$(\cdot)$ denotes the diagonal matrix. This diagonal bias emphasizes intrinsic change intensity at each spatial location and acts as a structured modulation term for subsequent attention logits.

Then, bidirectional CTA is computed by alternately using each temporal feature as the query and the other as key and value. The bias $\mathbf{B}^{(d)}$ is added to the attention logits in each direction to emphasize spatial locations with strong temporal discrepancies. Formally

$$\mathbf{A}_{T_1\to T_2} = \text{Softmax}\left(\frac{\mathbf{Q}_{T_1}\mathbf{K}^{\top}_{T_2}}{\sqrt{d}} + \mathbf{B}^{(T_1\to T_2)}\right) \quad (5)$$

$$\mathbf{A}_{T_2\to T_1} = \text{Softmax}\left(\frac{\mathbf{Q}_{T_2}\mathbf{K}^{\top}_{T_1}}{\sqrt{d}} + \mathbf{B}^{(T_2\to T_1)}\right) \quad (6)$$

where $\mathbf{Q}_{T_i}$ and $\mathbf{K}_{T_i}$ are the projected queries and keys, d is the key dimension, and $\mathbf{B}^{(d)}$ explicitly biases attention toward positions with higher direction-aware temporal discrepancies.

To further emphasize direction-aware temporal discrepancies, we combine the two directional attention responses via a differential operation

$$\mathbf{F}^{O}_{T_1\to T_2} = \left(\mathbf{A}_{T_1\to T_2} - \lambda\mathbf{A}_{T_2\to T_1}\right)\mathbf{V}_{T_2} \quad (7)$$

$$\mathbf{F}^{O}_{T_2\to T_1} = \left(\mathbf{A}_{T_2\to T_1} - \lambda\mathbf{A}_{T_1\to T_2}\right)\mathbf{V}_{T_1} \quad (8)$$

where $\mathbf{V}_{T_i}$ denotes the projection of the visual feature and λ is a balancing coefficient. By contrasting the two directional attentions, invariant responses are attenuated while asymmetric discrepancy cues are highlighted, without discarding contextual information entirely.

Finally, the resulting features are adaptively fused with the original bitemporal representations through a gated mechanism, yielding temporally enhanced features $\tilde{\mathbf{F}}^{l}_{T_1}$ and $\tilde{\mathbf{F}}^{l}_{T_2}$. These representations provide robust, discrepancy-aware change cues and are subsequently fed into the HASD module for coarse-to-fine semantic modeling.

C. Hierarchical Adaptive Semantic Disentanglement

To achieve comprehensive and fine-grained semantic understanding of bitemporal RSIs, we propose a HASD network that explicitly separates the modeling of changed and unchanged image pairs, as illustrated in Fig. 4. This design follows a coarse-to-fine strategy: first, coarse-grained image-level routing determines the existence of changes, and then fine-grained token-level routing determines the activated expert groups to capture diverse change semantics.

1) Coarse-Grained Image-Level Routing: At the coarse level, we design a difference-driven image-level routing mechanism to explicitly model the existence of changes in bitemporal image pairs.

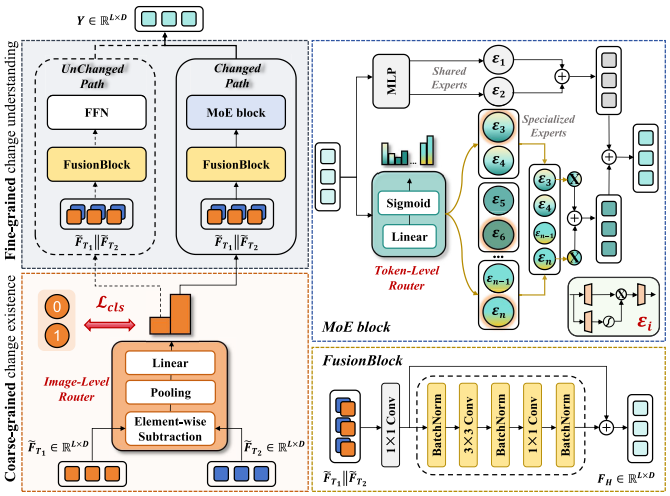


Fig. 4. Proposed hierarchical semantic disentanglement mechanism provides a structured perspective for RSICC by decomposing the task into two sequential stages: coarse-grained change existence perception driven by an image-level router and fine-grained change semantic understanding driven by a token-level router.

Given the enhanced features $\tilde{\mathbf{F}}_{T_1}$ and $\tilde{\mathbf{F}}_{T_2}$ from BDAM, we first compute their difference and apply global average pooling (GAP) to obtain a compact image-level representation

$$\hat{\mathbf{Z}} = \text{GAP}\left(\tilde{\mathbf{F}}_{T_2} - \tilde{\mathbf{F}}_{T_1}\right)\mathbf{W}_g \quad (9)$$

where $\mathbf{W}_g \in \mathbb{R}^{D\times E}$ is the learnable routing matrix mapping the D-dimensional feature to E expert paths (in our case, $E = 2$ corresponding to changed and unchanged paths).

The routing logits $\hat{\mathbf{Z}}$ are then converted to probabilities via softmax, and the most relevant expert path is selected using a top-1 routing strategy

$$(\mathcal{E}_k, \boldsymbol{\alpha}_k) = \text{Top-k}\left(\text{Softmax}\left(\hat{\mathbf{Z}}\right)\right), \qquad \sum_{e\in\mathcal{E}_k}\alpha_e = 1 \quad (10)$$

where $\mathcal{E}_k$ denotes the selected processing path index and $\boldsymbol{\alpha}_k$ contains the corresponding normalized routing weights. Specifically, $\mathcal{E}_k = 1$ indicates assignment to the changed-path expert, while $\mathcal{E}_k = 0$ corresponds to the unchanged-path expert.

To ensure semantically consistent and stable routing decisions, the predicted routing path is supervised with a cross-entropy classification loss

$$\mathcal{L}_{\text{cls}} = \text{CE}\left(\hat{\mathbf{Z}}, Z\right) \quad (11)$$

where $Z \in 0, 1$ denotes the ground-truth change label for the image pair. This coarse-grained routing mechanism effectively integrates change existence judgment into the network, providing a reliable foundation for the subsequent fine-grained token-level semantic disentanglement.

2) Fine-Grained Token-Level Disentanglement: Once the expert path is determined by the coarse-grained image-level routing, the enhanced bitemporal features are first aggregated via a lightweight FusionBlock equipped with convolutional operations to preserve spatio-temporal contextual information

$$\mathbf{F}_H = \text{FusionBlock}\left(\tilde{\mathbf{F}}_{T_1} \,\|\, \tilde{\mathbf{F}}_{T_2}\right) \quad (12)$$

where $\|$ denotes the concatenation along the feature dimension.

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For unchanged image pairs, the fused representations mainly encode temporally invariant semantics. Such representations can be effectively modeled using a simple feed-forward network (FFN).

In contrast, for changed image pairs, semantic changes are often complex and highly heterogeneous across tokens. A single transformation is therefore insufficient to model these diverse change patterns. To address this challenge, we introduce an MoE-based block that enables adaptive, tokenwise semantic modeling, allowing different experts to specialize in distinct types of fine-grained change semantics. Given the fused features $\mathbf{F}_H$, we perform token-level routing to dynamically assign each token to a small subset of specialized experts. Specifically, a lightweight gating module computes expert scores for each token

$$\mathbf{S}_{\text{expert}} = \sigma\left(\mathbf{F}_H \mathbf{W}_c\right) \tag{13}$$

where $\mathbf{S}_{\text{expert}} \in \mathbb{R}^{L\times E}$ denotes the expert scores for all L tokens over E experts, $\sigma(\cdot)$ is the sigmoid activation, and $\mathbf{W}_c \in \mathbb{R}^{D\times E}$ is the learnable routing matrix. In our design, $E = 8$ specialized experts are employed.

To mitigate uneven expert activation and encourage balanced utilization of specialized experts, we adopt a group-constrained routing [66] strategy. Specifically, expert relevance scores are first aggregated at the group level by applying max pooling over experts within each group

$$\mathbf{S}_{\text{group}} = \text{MaxPool}_{\text{group}}\left(\mathbf{S}_{\text{expert}}\right) \tag{14}$$

where each group score $\mathbf{S}_{\text{group}}$ reflects the strongest expert response in that group, serving as a proxy for the group's relevance to the current token.

Based on the resulting group-level scores, we select the top-k_g most relevant expert groups for each token, thereby constraining the subsequent expert selection to a compact and semantically coherent subset of experts

$$\mathcal{G}_{k_g}(n) = \text{Top-k}_{\text{g}}\left(\mathbf{S}_{\text{group}}(n,:)\right). \tag{15}$$

Then, expert-level sparse routing is conducted conditioned on the group-level selection $\mathcal{G}_{k_g}(n)$. For each token n, routing is confined to experts within the selected groups, producing the group-filtered expert scores $\tilde{\mathbf{S}}_{\text{expert}}$. The top-$k$ experts are subsequently activated according to their routing scores

$$\left(\mathcal{E}_k(n), \alpha_k(n)\right) = \text{Top-k}\left(\tilde{\mathbf{S}}_{\text{expert}}(n,:)\right), \quad \sum_{e\in\mathcal{E}_k(n)} \alpha_e = 1. \tag{16}$$

The outputs of the activated specialized experts are aggregated via a weighted sum according to their routing scores α_e, ensuring that each token integrates information from the most relevant experts. This aggregated representation is then combined with the output of the shared experts $f_s(\cdot)$, which capture global semantic information common to all tokens, yielding the final token representation

$$\mathbf{Y} = \sum_{e\in\mathcal{E}_k} \alpha_e f_e\left(\mathbf{F}_H\right) + f_{\text{s}}\left(\mathbf{F}_H\right) \tag{17}$$

where $f_i(\cdot)$ denotes the specialized expert implemented with a SwiGLU [67] FFN, with $i \in \{e, s\}$.

D. Multistage Curriculum Learning Strategy

The proposed model is trained in a fully supervised manner, where the primary objective is bitemporal change captioning. Given a predicted word distribution and the corresponding ground-truth caption, we optimize the model using the standard cross-entropy loss

$$\mathcal{L}_{\text{cap}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{v=1}^{V} y_n^{(v)} \log p_n^{(v)} \tag{18}$$

where N is the total number of words, V is the vocabulary size, $y_n^{(v)}$ denotes the one-hot ground-truth indicator at position n, and $p_n^{(v)}$ is the predicted probability of the vth word.

In addition to caption generation, we introduce an auxiliary image-level classification objective [see (11)] to supervise the coarse-grained routing in HASD. This auxiliary task serves a supportive role by enforcing semantically consistent expert-path selection, rather than acting as an independent prediction objective. However, we observe that directly jointly optimizing the captioning loss and the routing classification loss from scratch often leads to unstable training performance.

To address this issue, we adopt a multistage curriculum learning strategy that gradually introduces auxiliary supervision. Specifically, during the warm-up stage, gradients from the image-level routing branch are blocked by detaching the routing classification logits. This allows the model to first focus on learning robust visual-linguistic alignment and stable caption generation without interference from premature routing supervision. After the warm-up stage, the auxiliary routing classification loss is progressively activated using a cosine-based ramp-up schedule over the remaining epochs. This smooth transition prevents abrupt optimization shifts and enables the routing mechanism to adapt in accordance with increasingly reliable visual and linguistic representations. The overall training objective is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cap}} + \lambda\,\alpha(e)\,\mathcal{L}_{\text{cls}} \tag{19}$$

where $\mathcal{L}_{\text{cls}}$ denotes the cross-entropy loss for image-level change classification, λ is the balancing coefficient set to 0.8, and $\alpha(e)$ is the epoch-dependent cosine ramp-up factor.

The proposed multistage curriculum learning strategy effectively stabilizes training while ensuring that the auxiliary routing supervision is introduced in a controlled and semantically meaningful manner, ultimately leading to more reliable hierarchical semantic disentanglement and improved captioning performance.

IV. Experiments

A. Datasets

We evaluate our model HiSem on two widely used large benchmark datasets, LEVIR-CC [17] and WHU-CDC [68], for RSICC. LEVIR-CC [17] is a large-scale benchmark for RSICC, containing 10 077 bitemporal image pairs (5038 exhibiting semantic changes and 5039 unchanged) annotated with 50 385 human-annotated captions. The maximum caption length is 39 words, with an average sentence length of 7.99 words. All images are collected from 20 geographically

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TABLE I
PERFORMANCE COMPARISON OF OUR PROPOSED METHOD WITH PREVIOUS STATE-OF-THE-ART METHODS ON THE LEVIR-CC DATASET. THE BOLDED RESULTS ARE THE BEST

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
RSICCFormer [17] [TGRS'22]	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.66
Chg2Cap [1] [TIP'23]	86.14	78.08	70.66	64.39	40.03	75.12	136.61	79.04
PSNet [56] [IGARSS'23]	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
CTMTNet [68] [TGRS'24]	85.95	77.99	70.74	64.69	39.49	74.54	134.94	78.42
Sen [16] [TGRS'24]	85.10	77.05	70.01	64.09	39.59	74.57	136.02	78.57
SFT [18] [JSTARS'24]	84.56	75.87	68.64	62.87	39.93	74.69	137.05	78.64
Pix4Cap [48] [IGARSS'24]	85.56	77.08	69.79	63.78	39.96	75.12	136.76	78.91
RSCaMa [45] [GRSL'24]	85.79	77.99	71.04	65.24	39.91	75.24	136.56	79.24
Diffusion-RSCC [69] [TGRS'25]	-	-	-	60.90	37.80	71.50	125.60	73.95
Mask Approx Net [70] [TGRS'25]	85.90	77.12	70.72	64.32	39.91	75.67	137.71	79.40
Prompt-CC [2] [TGRS'23]	83.66	75.73	69.10	63.54	38.82	73.72	136.44	78.13
Chareption [64] [PRCV'24]	83.82	75.39	68.37	62.53	40.38	74.72	137.83	77.58
KCFI [21] [TIP'25]	86.34	77.31	70.89	65.30	39.42	75.47	138.25	79.61
HiSem (Ours)	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**

distinct regions across Texas, USA, with a temporal range spanning from 5 to 15 years. Each image is 256 × 256 pixels in size with a spatial resolution of 0.5 m/pixel.

WHU-CDC [68] comprises 7434 high-resolution bitemporal image pairs, annotated with a total of 37 170 descriptive captions. Each image is 256 × 256 pixels in size with a spatial resolution of 0.075 m/pixel. The dataset maintains a balanced ratio of changed and unchanged pairs. Caption annotations range from 3 to 24 words in length, with a total vocabulary of 327 unique words.

B. Evaluation Metrics

To rigorously assess the quality of the generated captions, we employ four widely used evaluation metrics, including BLEU-N (N = 1, 2, 3, 4), ROUGE$_L$, METEOR, and CIDEr-D. BLEU-N evaluates the precision of generated captions by comparing their n-grams with reference sentences, whereas ROUGE$_L$ focuses on recall by measuring the longest common subsequences between predictions and references. METEOR integrates both precision and recall while accounting for synonyms and stemming variations. CIDEr-D evaluates both the accuracy and diversity of generated captions by comparing their word frequency patterns with those of multiple reference captions, emphasizing consistency with the consensus across all references. Higher values on these metrics reflect closer agreement with the reference captions, indicating superior quality of the generated sentences. In addition, following previous methods, we adopt an average metric S_m^* as follows:

$$S_m^* = \frac{1}{4}(\text{BLEU-4} + \text{ROUGE}_\text{L} + \text{METEOR} + \text{CIDEr-D}). \tag{20}$$

C. Implementation Details

All models are implemented in PyTorch and trained on an NVIDIA GTX 4090 GPU. CLIP-ViT-B/16 [29] is adopted as the visual encoder, while a Transformer-based decoder is employed for language generation. Model optimization is performed using the Adam optimizer with an initial learning rate of 1×10^{-4}. The model is trained for 50 epochs with a batch size of 64. During inference, beam search is applied with a beam size of 1. The proposed BDAM module comprises three stacked layers, whereas the HASD module contains a single layer. The Transformer decoder is configured with one layer, and the embedding dimension is set to 768. To ensure statistical reliability, all experimental results are reported as the average over five independent runs.

D. Comparison to State of the Art

We compare the proposed method with representative state-of-the-art RSICC approaches on the LEVIR-CC and WHU-CDC benchmarks. Most of these methods generally follow an encoder–decoder framework, in which the encoder primarily aims to perceive and encode discriminative visual changes from bitemporal images, and the decoder subsequently maps these change-aware representations to natural language. To provide a clearer comparison, we categorize these methods according to their caption generation paradigms into three groups: Transformer-based autoregressive decoders, LLM-enhanced decoders, and alternative sequence modeling paradigms. Transformer-based autoregressive decoders constitute the mainstream paradigm in RSICC, including RSICCFormer [17], PSNet [56], CTMTNet [68], Sen [16], SFT [18], Pix4Cap [48], Chg2Cap [1], and Mask Approx Net [70]. These methods generally employ attention-based autoregressive decoding to generate change descriptions conditioned on encoded bitemporal visual representations. LLM-enhanced decoders include Prompt-CC [2], Chareption [64], and KCFI [21], which exploit pretrained LLMs to enhance change caption generation through visual-language alignment and instruction-guided generation. Other alternative sequence modeling paradigms have also emerged in RSICC. Diffusion-RSCC [69] formulates change caption generation as a progressive denoising process instead of conventional autoregressive generation, whereas RSCaMa [45] adopts a

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TABLE II
PERFORMANCE COMPARISON OF OUR PROPOSED METHOD WITH PREVIOUS STATE-OF-THE-ART METHODS ON THE WHU-CDC DATASET. THE BOLDED RESULTS ARE THE BEST

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
RSICCFormer [17] [TGRS'22]	80.05	74.24	69.61	66.54	42.65	73.91	133.44	79.14
Chg2Cap [1] [TIP'23]	78.93	72.64	67.20	62.71	41.46	77.95	144.18	81.58
PSNet [56] [IGARSS'23]	81.26	73.25	65.78	60.32	36.97	71.60	130.52	74.85
Sen [16] [TGRS'24]	80.60	74.64	67.69	61.97	36.76	71.70	133.57	76.00
SFT [18] [JSTARS'24]	81.17	72.90	66.06	60.27	37.34	72.63	134.64	76.22
Prompt-CC [2] [TGRS'23]	81.12	73.96	67.22	61.45	36.99	71.88	134.50	76.21
Diffusion-RSCC [69] [TGRS'25]	75.32	70.15	66.40	63.76	40.18	73.80	127.96	76.43
Mask Approx Net [70] [TGRS'25]	81.34	75.68	71.16	67.73	43.89	75.41	135.31	80.59
CTMTNet [68] [TGRS'24]	83.56	77.66	72.76	69.00	45.39	79.23	149.40	85.76
HiSem (Ours)	**87.49**	**82.89**	**79.17**	**76.52**	**48.77**	**82.00**	**158.35**	**91.41**

Mamba-based architecture to capture long-range dependencies during caption generation.

As reported in Tables I and II, HiSem consistently achieves state-of-the-art performance across all evaluation metrics on both datasets. On the LEVIR-CC dataset, HiSem surpasses the previous best-performing method KCFI with consistent gains. More notably, on the WHU-CDC dataset, HiSem achieves substantial improvements over previous methods (+7.52% on BLEU-4, +3.38% on METEOR, +2.77% on ROUGE$_L$, +8.95% on CIDEr-D, and +5.65% on S_m^*), indicating stronger capability in generating detailed and semantically faithful change descriptions. These performance gains reveal a key limitation shared by many existing approaches: they implicitly treat changed and unchanged samples at a uniform semantic granularity, assuming that all bitemporal image pairs require equally rich semantic modeling. In contrast, our method explicitly decouples the semantic representations of these two categories, enabling differentiated processing. The proposed BDAM module introduces a change-aware bidirectional differential attention mechanism that suppresses irrelevant background responses while selectively amplifying discriminative change cues. Building upon this, the HASD module further enables adaptive semantic routing by explicitly disentangling changed and unchanged samples into distinct processing paths. Through hierarchical semantic modeling, MoE-based experts can focus on fine-grained change semantics without interference from unchanged samples while maintaining compact and invariant representations for stable scenes. This design leads to more semantically consistent and informative captions, as evidenced by the pronounced gains in the evaluation metrics, particularly on WHU-CDC.

Overall, these results demonstrate that our proposed hierarchical semantic disentanglement idea provides a more effective modeling perspective for RSICC, especially in scenarios involving complex and heterogeneous change distributions, validating the robustness and generalization capability of the proposed HiSem method.

E. Qualitative Results

1) Qualitative Comparison Results: Fig. 5 presents qualitative comparisons of captioning results between the proposed HiSem method and several state-of-the-art approaches, including Pix4Cap [48], Prompt-CC [2], Chg2Cap [1], and RSCaMa [45], on the LEVIR-CC dataset. In the figure, red highlights indicate more accurate and detailed descriptions, whereas green highlights denote incorrect descriptions.

In the first row, the four image pairs involve both appearance and disappearance change patterns. Our proposed HiSem accurately captures both change states within the same pair, whereas other methods tend to miss one of the changes or produce inaccurate descriptions. The second row features complex scenarios with multiple objects and interobject spatial relationships. Our HiSem generates comprehensive and fine-grained descriptions, correctly reflecting object locations, relationships, and multiple changed objects. For instance, in the first image pair of the second row, our method generates a detailed description of locations (e.g., "A house appears in the woods in the top left corner of the scene."), whereas Pix4Cap only produces a incorrect description (e.g., "A house appears in the woods at bottom."). In the last row, involving subtle changes and irrelevant variations, HiSem still successfully identifies fine-grained semantic changes, demonstrating robustness under challenging conditions. Overall, these qualitative captioning results demonstrate that our proposed method can accurately and comprehensively capture the fine-grained semantic information while suppressing irrelevant variations to enable more precise, detailed, and semantically faithful change captions.

2) Feature Visualization: To further validate the effectiveness of the BDAM module in extracting discriminative representations, we visualize feature maps for image pairs from the LEVIR-CC and WHU-CDC datasets before and after applying BDAM (see Fig. 6). For the LEVIR-CC examples [see Fig. 6(a)], which contain subtle changes, BDAM introduces discrepancy-guided cross-temporal interactions, producing consistent and accurate attention responses across both temporal observations. Similarly, for the WHU-CDC examples [see Fig. 6(b)], where the before-to-after transition involves both appearance and disappearance changes, the feature maps show that BDAM simultaneously emphasizes multiple types of true changes while suppressing irrelevant variations. These observations demonstrate that BDAM

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Fig. 5. Captioning results generated by Pix4Cap, Prompt-CC, Chg2Cap, RSCaMa, and our HiSem on the LEVIR-CC dataset. GT denotes one of the five ground-truth captions. Text highlighted in red corresponds to more accurate and detailed descriptions, whereas text highlighted in green represents incorrect descriptions.

effectively guides directional cross-temporal interactions, yielding more discriminative representations that support fine-grained and robust change perception.

F. Ablation Studies

1) Effectiveness of BDAM and HASD: Table III presents the ablation results of the proposed BDAM and HASD modules on the LEVIR-CC dataset. Starting from the baseline model, incorporating the BDAM module yields consistent improvements across all evaluation metrics, achieving gains of +1.10% on BLEU-4, +5.63% on CIDEr-D, and +2.34% on S_m^*. These results demonstrate that bidirectional discrepancy-guided attention effectively enhances discriminative bitemporal representations and improves the localization of change regions. When HASD is further integrated, the model achieves the best performance across all metrics. Compared with the baseline, the full model improves BLEU-4 by +1.50%, CIDEr-D by +7.02%, and S_m^* by +3.05%. These results indicate that adaptive semantic routing and expert specialization effectively reduce semantic interference between changed and unchanged samples to obtain high-quality change-aware representations, enabling more accurate, detailed, and semantically consistent change descriptions.

2) Detailed Component Effectiveness Analysis of BDAM: Table IV investigates the contribution of each sublayer within

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TABLE III
ABLATION STUDIES ON THE BDAM AND HASD MODULES ON THE LEVIR-CC DATASET. THE BOLDED RESULTS ARE THE BEST

Method	BDAM	HASD	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
Baseline	✗	✗	84.51	76.73	69.89	64.32	38.79	73.68	131.84	77.16
–	✔	✗	86.07	78.16	71.17	65.42	39.94	75.14	137.47	79.50
–	✗	✔	83.31	75.43	68.40	62.70	38.02	72.33	129.09	75.54
HiSem (Ours)	✔	✔	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**

TABLE IV
ABLATION STUDIES ON THE COMPONENTS OF THE BDAM MODULE ON THE LEVIR-CC DATASET, ANALYZING THE EFFECTS OF THE DFC SUBLAYER AND BDCI SUBLAYER

DFC Sublayer	BDCI Sublayer	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
✗	✗	83.31	75.43	68.40	62.70	38.02	72.33	129.09	75.54
✔	✗	83.73	75.57	68.44	62.65	38.25	72.49	130.13	75.88
✗	✔	84.51	75.90	68.74	62.97	39.51	74.50	135.35	78.08
✔	✔	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**

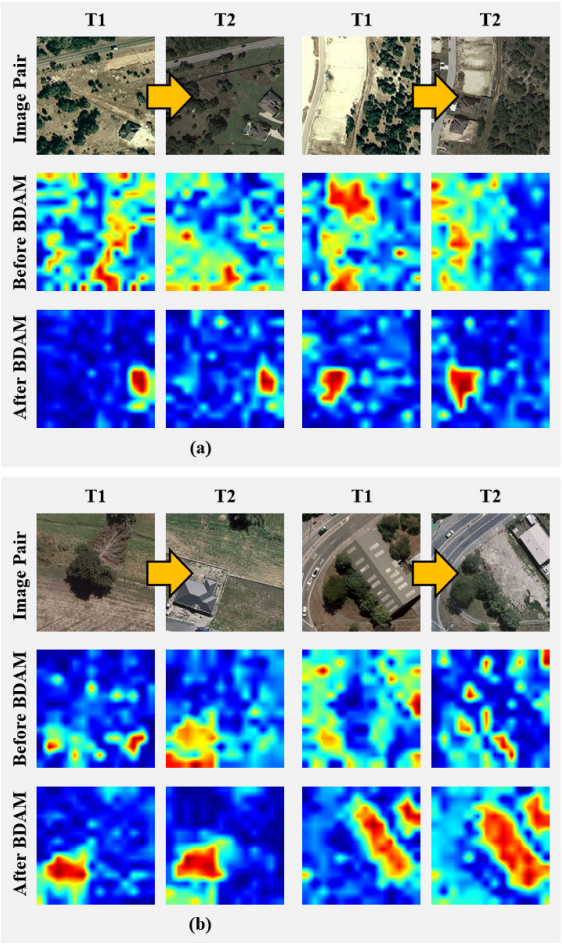


Fig. 6. Visualization of features before and after BDAM module processing. (a) LEVIR-CC dataset. (b) WHU-CDC dataset.

the BDAM module, containing the DFC and the BDCI. The results reveal a compelling synergistic effect between the two sublayers to facilitate the modeling of synergistic spatial and temporal information. Specifically, while the independent integration of the DFC sublayer yields relatively stable performance, it serves as a critical representation foundation by refining the spatial representation of bitemporal features. The modest gains observed with DFC alone suggest that purely spatial enhancement, without effective temporal reasoning, is insufficient for complex change recognition. On the other hand, the BDCI sublayer provides a more significant performance boost (e.g., +6.26% in CIDEr-D) by utilizing bidirectional discrepancies as structural priors to guide CTA. This validates the importance of explicit discrepancy modeling in capturing authentic changes.

The most crucial observation is the significant performance leap achieved through the joint integration of both sublayers. The full BDAM configuration significantly outperforms the baseline by +3.12% in BLEU-4 and a substantial +9.77% in CIDEr-D, which exceeds the cumulative gains of the two individual sublayers. This significant improvement demonstrates that DFC effectively "recalibrates" the feature space, allowing the subsequent BDCI to more precisely extract semantic changes. These results strongly validate our design philosophy: the sequential transition from spatial feature conditioning to discrepancy-guided temporal interaction is not merely additive but fundamentally essential for decoding the intricate dynamics of remote sensing imagery.

3) Effectiveness of Hierarchical Semantic Disentanglement: Table V evaluates the core design philosophy of the HASD module: the decoupling of bitemporal change perception into a hierarchical routing and task-tailored modeling process. In a monolithic architecture (Rows 2 and 3 in Table V) where no image-level routing is applied, replacing standard FFNs with MoE consistently yields performance gains across all metrics (e.g., +1.65% in CIDEr-D). This confirms that MoE's sparse activation mechanism possesses a superior inductive bias for capturing the heterogeneous semantic changes

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TABLE V

ABLATION STUDIES ON THE COMPONENTS OF THE HASD MODULE ON THE LEVIR-CC DATASET, ANALYZING THE EFFECT OF INTRODUCING COARSE-GRAINED IMAGE-LEVEL ROUTING UNDER DIFFERENT SEMANTIC MODELING ARCHITECTURES (FFN VERSUS MOE). $\mathcal{R}_{\mathrm{I}}$ DENOTES THE COARSE-GRAINED IMAGE-LEVEL ROUTING CONFIGURATION, WHERE "HARD" INDICATES TOP-1 PATH ASSIGNMENT AND "SOFT" DENOTES CONFIDENCE-WEIGHTED FEATURE PATH AGGREGATION. $\mathcal{L}_{\mathrm{cls}}$ DENOTES THE IMAGE-LEVEL CLASSIFICATION LOSS

$\mathcal{R}_{\mathrm{I}}$	$\mathcal{L}_{\mathrm{cls}}$	Changed Path		Unchanged Path		BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
		FFN	MoE	FFN	MoE								
–	✗	✗	✗	✗	✗	86.07	78.16	71.17	65.42	39.94	75.14	137.47	79.50
–	✗	✔	✗	✔	✗	86.17	78.20	71.02	64.99	40.00	75.18	137.00	79.29
–	✗	✗	✔	✗	✔	86.38	78.45	71.34	65.38	40.37	75.70	138.65	80.02
Hard	✔	✔	✗	✔	✗	85.42	77.59	70.75	65.15	39.90	75.00	136.42	79.12
Hard	✔	✗	✔	✗	✔	85.72	77.90	70.87	65.10	39.62	75.02	135.87	78.90
Hard	✔	✗	✔	✔	✗	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**
Soft	✔	✗	✔	✔	✗	85.69	77.94	70.98	65.25	39.62	74.88	135.17	78.73
Hard	✗	✗	✔	✔	✗	85.89	78.23	71.18	65.37	39.80	75.05	136.32	79.14

TABLE VI

PERFORMANCE ANALYSIS OF OUR SEMANTIC DISENTANGLEMENT MECHANISM ON THE LEVIR-CC DATASET. WE COMPARE TWO INFERENCE SETTINGS: *Routing (w. Pre)*, WHICH RELIES ON PREDICTED CHANGE EXISTENCE, AND *Routing (w. GT)*, WHICH EMPLOYS GROUND-TRUTH LABELS TO ELIMINATE ROUTING UNCERTAINTY. NOTABLY, WE INTRODUCE A NORMALIZED PERFORMANCE GAIN DEFINED AS ρ, WHICH MEASURES THE CONVERSION EFFICIENCY FROM ROUTING IMPROVEMENTS TO LANGUAGE GENERATION GAINS

Evaluation Scope	Method	Acc_{router}	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
Unchanged image pairs	Routing *(w. Pre.)*	96.91%	97.22	96.80	96.57	96.41	75.75	97.55	–	–
	Routing *(w. GT)*	100%	100.00	100.00	100.00	100.00	100.00	100.00	–	–
	Improvement (Δ)	**+3.09%**	**+2.78**	**+3.20**	**+3.43**	**+3.59**	**+24.25**	**+2.45**	–	–
	$\rho = \Delta Score\ /\Delta Acc_{\mathrm{router}}$	–	**0.90**	**1.04**	**1.11**	**1.16**	**7.85**	**0.79**	–	–
Changed image pairs	Routing *(w. Pre)*	91.06%	77.89	63.94	51.15	40.41	25.92	53.96	65.12	46.35
	Routing *(w. GT)*	100%	79.69	65.96	52.95	41.78	27.05	56.32	69.54	48.67
	Improvement (Δ)	**+8.94%**	**+1.80**	**+2.02**	**+1.80**	**+1.37**	**+1.13**	**+2.36**	**+4.42**	**+2.32**
	$\rho = \Delta Score\ /\Delta Acc_{\mathrm{router}}$	–	**0.20**	**0.23**	**0.20**	**0.15**	**0.13**	**0.26**	**0.49**	**0.26**
All image pairs	Routing *(w. Pre)*	93.99%	86.60	78.74	71.73	65.82	40.39	75.77	138.86	80.21
	Routing *(w. GT)*	100%	88.53	80.91	73.82	67.69	41.89	78.17	144.63	83.09
	Improvement (Δ)	**+6.01%**	**+1.93**	**+2.17**	**+2.09**	**+1.87**	**+1.50**	**+2.40**	**+5.77**	**+2.88**
	$\rho = \Delta Score\ /\Delta Acc_{\mathrm{router}}$	–	**0.32**	**0.36**	**0.35**	**0.31**	**0.25**	**0.40**	**0.96**	**0.48**

inherent in remote sensing temporal imagery compared with static FFNs. A pivotal observation arises when introducing the coarse-grained image-level routing. Interestingly, maintaining a *symmetric* architecture for both paths (Rows 4 and 5) leads to a performance bottleneck or even degradation compared with their monolithic counterparts. This phenomenon suggests that once the router explicitly disentangles the "changed" and "unchanged" semantic streams, employing identical representational capacity for both becomes suboptimal. Unchanged samples, characterized by high semantic homogeneity, do not require the high-dimensional expert space of an MoE; forcing them through a complex routing-based ensemble may introduce unnecessary noise or over-fitting.

The optimal performance is achieved in Row 6, where we implement an *asymmetric* strategy with *hard* Top-1 routing: a specialized MoE for the high-entropy "changed" path and a parsimonious FFN for the low-entropy "unchanged" path. This configuration achieves the best overall results among the evaluated variants, with CIDEr-D and S_m^* reaching 138.86 and 80.21, respectively. We further investigate a *soft* routing variant in Row 7, in which the outputs of the two semantic paths are aggregated using the router confidence scores as weights. The performance degradation relative to Row 6 suggests that explicit Top-1 assignment is more compatible with the proposed hierarchical semantic disentanglement framework than confidence-weighted aggregation. This observation implies that *soft* routing may introduce ambiguity in branch-specific semantic allocation, thereby weakening semantic specialization and reducing the effectiveness of disentangled heterogeneous semantic modeling. To isolate the effect of the classification loss $\mathcal{L}_{\mathrm{cls}}$, we compare Row 6 and Row 8 under an otherwise identical *asymmetric hard*-routing architecture, where Row 8 removes $\mathcal{L}_{\mathrm{cls}}$ during training. The consistent performance degradation across evaluation metrics suggests that $\mathcal{L}_{\mathrm{cls}}$ facilitates more reliable image-level semantic routing, which in turn improves hierarchical semantic disentanglement and downstream change caption generation. These results empirically validate our hypothesis: the RSICC task is best addressed by first explicitly bifurcating the data stream and then aligning the model capacity with the intrinsic

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TABLE VII

PERFORMANCE ANALYSIS OF OUR SEMANTIC DISENTANGLEMENT MECHANISM ON THE WHU-CDC DATASET. WE COMPARE TWO INFERENCE SETTINGS: *Routing (w. Pre)*, WHICH RELIES ON PREDICTED CHANGE EXISTENCE, AND *Routing (w. GT)*, WHICH EMPLOYS GROUND-TRUTH LABELS TO ELIMINATE ROUTING UNCERTAINTY. NOTABLY, WE INTRODUCE A NORMALIZED PERFORMANCE GAIN DEFINED AS ρ, WHICH MEASURES THE CONVERSION EFFICIENCY FROM ROUTING IMPROVEMENTS TO LANGUAGE GENERATION GAINS

Evaluation Scope	Method	Acc_{router}	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
Unchanged image pairs	Routing *(w. Pre)*	97.40%	98.07	97.85	97.77	97.74	77.94	97.89	–	–
	Routing *(w. GT)*	100%	100.00	100.00	100.00	100.00	100.00	100.00	–	–
	Improvement (Δ)	**+2.60%**	**+1.93**	**+2.15**	**+2.23**	**+2.26**	**+22.06**	**+2.11**	–	–
	$\rho = \Delta Score\ /\Delta Acc_{\text{router}}$	–	**0.74**	**0.83**	**0.86**	**0.87**	**8.48**	**0.81**	–	–
Changed image pairs	Routing *(w. Pre)*	80.08%	65.51	50.92	37.89	28.49	22.45	47.78	58.98	39.42
	Routing *(w. GT)*	100%	70.97	57.42	43.80	33.15	25.12	53.66	71.49	45.86
	Improvement (Δ)	**+19.92%**	**+5.46**	**+6.50**	**+5.91**	**+4.66**	**+2.67**	**+5.88**	**+12.51**	**+6.44**
	$\rho = \Delta Score\ /\Delta Acc_{\text{router}}$	–	**0.27**	**0.33**	**0.30**	**0.23**	**0.13**	**0.30**	**0.63**	**0.32**
All image pairs	Routing *(w. Pre)*	91.91%	87.49	82.89	79.17	76.52	48.77	82.00	158.35	91.41
	Routing *(w. GT)*	100%	90.37	86.16	82.26	79.28	51.06	85.30	167.27	95.73
	Improvement (Δ)	**+8.09%**	**+2.88**	**+3.27**	**+3.09**	**+2.76**	**+2.29**	**+3.30**	**+8.92**	**+4.32**
	$\rho = \Delta Score\ /\Delta Acc_{\text{router}}$	–	**0.36**	**0.40**	**0.38**	**0.34**	**0.28**	**0.41**	**1.10**	**0.53**

semantic complexity of each subtask. This "divide-and-conquer" approach not only enhances descriptive precision for complex changes but also ensures structural efficiency for consistent scenes.

G. Insights Into Hierarchical Semantic Disentanglement

The proposed hierarchical semantic disentanglement mechanism provides a structured perspective for RSICC by decomposing the task into two sequential stages: coarse-grained change existence perception and fine-grained change semantic understanding. By explicitly separating changed and unchanged image pairs, this design alleviates semantic entanglement and enables a divide-and-conquer modeling strategy, where different components can be designed for distinct semantic objectives.

To further analyze the implications of this strategy, we conduct additional experiments on the LEVIR-CC and WHU-CDC datasets (see Tables VI and VII). We compare two inference settings: *Routing (w. Pre)*, which relies on predicted change existence, and *Routing (w. GT)*, which employs ground-truth labels to eliminate routing uncertainty, thereby representing the upper bound performance of the proposed framework. Notably, to quantify how routing accuracy translates into caption quality, we introduce a normalized performance gain defined as $\rho = \Delta\text{Score}/\Delta Acc_{\text{router}}$, which measures the conversion efficiency from routing improvements to language generation gains. Following the evaluation protocol of [17], CIDEr-D is omitted for unchanged image pairs in Tables VI and VII because the highly repetitive no-change captions result in limited discriminative capability under consensus-based evaluation.

1) Justification of Disentangling Change Existence and Semantic Understanding: A consistent observation across both datasets is that, for changed image pairs, substantial improvements in routing accuracy under ground-truth guidance (+8.94% on LEVIR-CC and +19.92% on WHU-CDC) lead to only moderate gains in captioning performance (+1.37% and +4.66% on BLEU-4, respectively), resulting in significantly lower normalized gains compared with unchanged samples. This phenomenon indicates that, while coarse-grained change existence perception can be improved relatively reliably, capturing the underlying fine-grained semantics of changes remains considerably more challenging. Such a discrepancy reveals an intrinsic property of RSICC: changed and unchanged image pairs exhibit fundamentally different levels of semantic complexity. Unchanged pairs primarily require coarse-grained semantic confirmation, whereas changed pairs demand detailed modeling of object attributes, spatial relationships, and temporal evolution. This inherent asymmetry justifies the necessity of explicitly disentangling their representations, as uniform modeling inevitably leads to semantic interference and suboptimal performance.

2) Routing Accuracy as a Bottleneck for Coarse-to-Fine Modeling: Across both datasets, we observe a consistent positive correlation between routing accuracy and captioning performance, suggesting that the quality of the routing signal plays a pivotal role in the overall framework. This finding identifies coarse-grained change existence perception as a critical bottleneck, emphasizing the importance of enhancing the discriminative capability of the BDAM module to provide reliable inputs for subsequent fine-grained semantic modeling.

3) From Coarse-Grained Routing to Fine-Grained Semantic Modeling: Despite its effectiveness, the relatively low conversion efficiency observed for changed samples indicates that accurately identifying change existence does not directly translate into equally strong improvements in fine-grained semantic description. In our framework, this gap is explicitly addressed by the HASD module, where an MoE-based design enables token-level modeling of diverse change patterns by assigning different experts to specialize in distinct semantic variations, thereby moving beyond binary routing toward fine-grained semantic understanding. Nevertheless, the remaining discrepancy suggests that modeling complex real-world changes remains challenging, and further improvements may

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TABLE VIII
COMPARISON OF DIFFERENT VISUAL BACKBONE PRETRAINING STRATEGIES ON THE LEVIR-CC DATASET

Vision Transformer		BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
ImageNet	ViT-B/32	85.00	76.87	69.91	64.22	39.42	74.28	135.35	78.32
	ViT-B/16	86.14	78.25	71.19	65.26	39.80	75.10	137.03	79.30
CLIP	ViT-B/32	85.93	78.13	71.10	65.26	39.85	75.09	136.56	79.19
	ViT-B/16	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**

TABLE IX
DEPTH SENSITIVITY ANALYSIS OF BDAM AND TRANSFORMER DECODER ON LEVIR-CC. N_{BDAM} AND N_{Dec} DENOTE THE NUMBER OF BDAM LAYERS AND TRANSFORMER DECODER LAYERS, RESPECTIVELY

N_{BDAM}	N_{Dec}	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
1	1	85.99	77.93	70.72	64.74	40.24	75.42	138.11	79.63
2	1	86.11	78.13	71.14	65.33	40.13	75.24	137.55	79.56
3	1	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**
3	2	85.94	78.01	70.86	64.91	39.72	74.79	136.31	78.93
3	3	86.07	77.96	70.94	65.19	40.24	75.46	137.64	79.63

TABLE X
SENSITIVITY ANALYSIS OF MOE ROUTING AND CAPACITY DESIGN FOR FINE-GRAINED CHANGE SEMANTIC MODELING ON THE LEVIR-CC DATASET

MoE Design Configurations					BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
$E_{\text{specialized}}$	n_{group}	Top-k_g	Top-k	E_{shared}								
8	4	2	1	2	86.10	78.06	71.04	65.21	40.15	75.43	138.24	79.76
8	4	2	2	2	**86.60**	**78.74**	**71.73**	**65.82**	**40.39**	**75.77**	**138.86**	**80.21**
8	2	1	2	2	86.10	78.27	71.36	65.55	40.02	75.19	137.52	79.57
8	4	1	2	2	85.93	78.17	71.28	66.11	39.68	75.05	137.22	79.52
6	3	2	2	2	86.32	78.32	71.19	65.23	40.31	75.55	138.04	79.78
10	5	2	2	2	86.02	78.20	71.20	65.42	39.95	75.31	136.67	79.34
8	4	2	2	1	86.18	78.24	71.15	65.27	40.07	75.29	138.12	79.69

be achieved by enhancing expert specialization and designing more expressive routing strategies to better align coarse-grained perception with high-fidelity semantic generation.

4) Implications for Future RSICC Research: Overall, the proposed mechanism introduces a clear separation between change existence perception and semantic understanding, providing a structured guideline for RSICC modeling. This divide-and-conquer strategy enables the design of specialized components for different semantic granularities, thereby reducing semantic confusion and improving descriptive accuracy in complex remote sensing scenarios.

H. Parametric Experiments and Analysis

1) Effect of Visual Backbone Architectures and Pretraining Strategies: Table VIII compares different ViT backbones under ImageNet and CLIP pretraining schemes. Across both pretraining settings, ViT-B/16 consistently outperforms ViT-B/32, suggesting that finer patch granularity is more effective for preserving subtle spatial semantics in bitemporal RSIs. Moreover, CLIP-pretrained backbones achieve consistently better performance than ImageNet-pretrained counterparts under the same architecture (e.g., CIDEr-D improves from 137.03 to 138.86 under ViT-B/16), indicating that vision-language pretraining produces semantically more transferable visual representations for downstream change caption generation. Overall, the superior performance of CLIP ViT-B/16 demonstrates that combining finer grained visual encoding with semantically aligned pretraining is beneficial for fine-grained change semantic modeling.

2) Depth Analysis of BDAM and the Caption Decoder: We vary the numbers of BDAM and Transformer decoder layers in Table IX, denoted as N_{BDAM} and N_{Dec}, respectively. The configuration with $N_{\text{BDAM}} = 3$ and $N_{\text{Dec}} = 1$ achieves the best overall performance, suggesting that a moderate BDAM depth is sufficient to progressively refine bitemporal representations through discrepancy-aware spatial modulation and bidirectional cross-temporal interaction. By contrast, deeper decoder configurations do not bring consistent improvements over the one-layer decoder, suggesting that further increasing

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TABLE XI
IMPACT OF BEAM SIZE ON INFERENCE PERFORMANCE ON THE LEVIR-CC DATASET

Beam Size	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	Inference Speed
1	**86.60**	78.74	71.73	65.82	**40.39**	**75.77**	**138.86**	15.5 fps
3	86.45	**78.96**	72.42	67.07	40.35	75.66	138.66	11.1 fps
5	86.26	78.90	**72.53**	**67.36**	40.38	75.64	138.62	10.2 fps

decoder depth provides limited benefit when the upstream modules already produce sufficiently discriminative features.

3) Sensitivity Analysis of MoE Routing and Capacity Design in HASD: Table X evaluates the sensitivity of HASD from three complementary perspectives: routing behavior, expert capacity, and shared expert capacity. Among all evaluated settings, the configuration with $E_{\text{specialized}} = 8$, $n_{\text{group}} = 4$, Top-$k_g = 2$, Top-$k = 2$, and $E_{\text{shared}} = 2$ achieves the best overall performance, suggesting that effective fine-grained change semantic modeling within HASD depends on balancing expert specialization and expert accessibility, rather than independently scaling individual MoE hyperparameters.

a) Routing behavior: Increasing Top-k from 1 to 2 consistently improves performance, suggesting that enabling each token to access multiple experts improves semantic expressiveness. At the group level, the comparison between Row 2 and Rows 3 and 4 suggests that increasing grouping granularity alone does not guarantee better performance. Under group-constrained routing, sufficient group activation is necessary to preserve expert accessibility and maintain sufficient candidate expert coverage.

b) Expert capacity: The sensitivity to expert capacity exhibits a nonmonotonic trend. To decouple expert capacity from grouping granularity, Row 2 and Rows 5 and 6 follow the same grouping rule, where each group contains two specialized experts and n_{group} is adjusted accordingly. The results indicate that increasing the number of specialized experts is beneficial only up to a moderate scale. Insufficient expert capacity limits the modeling of diverse change semantics, whereas an excessively large expert pool may over-partition the routing space and reduce expert utilization efficiency. In our setting, this balance is achieved when $E_{\text{specialized}} = 8$.

c) Shared expert capacity: Increasing E_{shared} from 1 to 2 consistently improves performance, suggesting that shared experts provide shared semantic representations that complement specialized expert routing. This shared pathway helps to alleviate excessive dependence on highly partitioned expert specialization and contributes to more robust semantic learning.

4) Beam Search Strategy for Inference: Table XI evaluates the effect of beam size during inference, where beam size = 1 corresponds to greedy decoding. Increasing the beam size from 1 to 5 leads to marginal improvements in BLEU-3 and BLEU-4, but does not yield consistent gains across other evaluation metrics. Meanwhile, inference speed decreases from 15.5 to 10.2 fps. These results suggest that expanding the decoding search space provides limited overall benefit under the current setting. Therefore, considering the tradeoff between captioning performance and inference efficiency, beam size = 1 is adopted as the default inference strategy.

V. CONCLUSION

In this article, we reveal a fundamental limitation in existing RSICC methods: the unified modeling of changed and unchanged image pairs overlooks their inherently different semantic complexities. To address this issue, we propose a HiSem, which explicitly decomposes the task into coarse-grained change existence perception and fine-grained change semantic understanding. Building upon this insight, the proposed BDAM module enhances discriminative change existence perception via discrepancy-aware cross-temporal interaction, while the HASD module enables adaptive and specialized modeling through hierarchical routing and expert-based semantic understanding. Extensive experiments demonstrate that HiSem consistently achieves superior performance across two benchmarks. More importantly, this work provides a structured perspective for RSICC by explicitly aligning model design with the intrinsic semantic heterogeneity of bitemporal scenes.

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Man Wang is currently pursuing the Ph.D. degree with the Research Center for Spatiotemporal Intelligence, College of Computer Science, Inner Mongolia University, Hohhot, China.

Her research interests include computer vision, remote sensing image processing, and multimodal remote sensing image understanding and intelligent interpretation.



Chenyang Liu received the B.S. degree from the Image Processing Center, School of Astronautics, Beihang University, Beijing, China, in 2021, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision, multimodal learning, and generative AI. For more information, see https://chen-yang-liu.github.io/



Wenjun Li is currently the Director of Inner Mongolia Data Center, National Beidou Navigation Location Service Data Center, and an Expert of China Mobile Communications Group Inner Mongolia Company Ltd., Hohhot, China. His main research interests include large-scale application of Beidou, satellite remote sensing image application, and unmanned aerial vehicle (UAV) monitoring.



Feng Ni is currently working with China Mobile Communications Group Inner Mongolia Company Ltd., Hohhot, China, where his research concentrates on big data and artificial intelligence. He has published over ten academic articles independently or collaboratively, and holds four granted patents as well as two registered software copyrights.



Bing Jia (Member, IEEE) received the Ph.D. degree from Jilin University, Changchun, China, in 2013.

She is currently a Professor with the College of Computer Science, Inner Mongolia University, Hohhot, China. Her research interests include indoor localization, crowdsourcing, wireless sensor networks, and mobile computing.



Baoqi Huang received the B.E. degree in computer science from Inner Mongolia University (IMU), Hohhot, China, in 2002, the M.S. degree in computer science from Peking University, Beijing, China, in 2005, and the Ph.D. degree in information engineering from Australian National University, Canberra, ACT, Australia, in 2012.

From May 2013 to April 2014, he worked as a Research Fellow with Nanyang Technological University, Singapore. From July 2018 to July 2019, he worked as a Visiting Professor with the University of Technology Sydney, Sydney, New South Wales, Australia. He is currently a Professor with the College of Computer Science, IMU. His research interests include smart sensing, wireless communications, and mobile computing.

Dr. Huang was a recipient of Chinese Government Award for Outstanding Chinese Students Abroad in 2011.



Riting Xia received the Ph.D. degree from Jilin University, Changchun, China, in 2024.

She is currently a Professor with the College of Computer Science and the Research Center for Spatiotemporal Intelligence, Inner Mongolia University, Hohhot, China. She has authored or co-authored more than 20 scientific articles in refereed journals and proceedings. Her research interests include remote sensing image processing and analysis, computer vision, graph data mining, and machine learning. For more information, see https://ccs.imu.edu.cn/info/1023/2411.htm



Zhenwei Shi (Senior Member, IEEE) is currently a Professor. He has authored or co-authored more than 300 scientific articles in refereed journals and proceedings. His research interests include remote sensing image processing and analysis, computer vision, pattern recognition, and machine learning.

Prof. Shi serves as an Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Pattern Recognition*, and *Infrared Physics and Technology*.

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